

Knot Too Early: The Causal Impact of Women’s Age at Marriage on Period Product Use

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February 16, 2026

Abstract

Despite significant progress in recent years, inadequate menstrual health and hygiene (MHH) remains a critical public health challenge in India, with approximately 30 percent of women aged 15–24 continuing to rely on cloth, rags, or nothing during menstruation. This paper examines whether delaying marriage can improve the menstrual health outcomes by estimating the causal effect of women’s age at marriage on their use of period products. Using nationally representative data from India’s National Family Health Survey (NFHS-5, 2019-21), we address endogeneity concerns by instrumenting age at marriage with age at menarche, a largely biologically determined factor that strongly predicts marriage timing in settings where daughters are married soon after attaining reproductive maturity. Our instrumental variable estimates reveal that delaying marriage by one year increases the probability of using period products by 7 percentage points (pp). This result is robust to alternative outcome specifications, sample restrictions, and falsification tests. We provide suggestive evidence that the effect operates through two key mechanisms: increased educational attainment and enhanced household decision-making power. Heterogeneity analysis reveals that effects are concentrated among socioeconomically disadvantaged groups, particularly women from backward castes and rural areas. These findings suggest that policies aimed at delaying child marriage, including stricter enforcement of existing laws and expansion of conditional cash transfer programs, may yield meaningful improvements in menstrual health outcomes, particularly for disadvantaged populations.

JEL Classifications: C36, I15, I18, J12, J16

Keywords: Menstrual Hygiene, Age at Marriage, Age at Menarche, Instrumental Variable, India

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1 Introduction

Globally, menstruation is a natural biological process experienced by women of reproductive age, yet inadequate menstrual health management remains a critical public health challenge in low- and middle-income countries. On any given day, more than 300 million women menstruate across the world, and approximately 1.8 billion individuals experience menstruation each month (Nature Human Behaviour, 2024; UNICEF, 2025). Despite its universality, an estimated half a billion women worldwide suffer from “period poverty”. It is defined as the inability to afford and access menstrual products, sanitation facilities, and the education needed to manage menstrual health with dignity (UN Women, 2025; World Bank Group, n.d.).

The consequences of period poverty extend far beyond physical discomfort. Inadequate menstrual health and hygiene (MHH) is associated with higher risks of reproductive and urinary tract infections (Babbar & Ojha, 2025; Bhusal, 2020; Das et al., 2015; Das et al., 2025), mental health complications (Babbar, 2025), and restrictive social norms that constrain women’s mobility, limit economic opportunities (Babbar et al., 2022; Singh et al., 2022), and reduce overall quality of life (Babbar et al., 2022; Sommer & Sahin, 2013). Moreover, inadequate MHH has also been shown to have a negative effect on girls’ and women’s educational participation (Oster & Thornton, 2011).

India presents an important case for examining menstrual health outcomes. Home to nearly 355 million menstruating women and one of the largest populations of adolescent girls, India has made great progress in menstrual health practices over the past decade (Chauhan et al., 2021). The share of women aged 15–24 who use modern period products rose sharply from only 15% in 2007–08 to 77.5% in 2019–21 (Babbar & Garikipati, 2023). However, this aggregate figure masks important gaps. Fewer than half of women in this age group (47.5%) report exclusive use of period products, while approximately 30% continue to rely on non period products such as cloth, rags, or nothing at all. These persistent gaps exist despite government interventions including complete Goods and Services Tax (GST) exemption on sanitary pads and the implementation of large-scale distribution programmes such as the Menstrual Hygiene Scheme, Kerala’s She-Pad Scheme, and Odisha’s Khushi Scheme.

Why do these gaps persist? The existing literature has identified multiple barriers to the adoption of period products, including affordability constraints, lack of awareness, and deeply entrenched cultural taboos surrounding menstruation (Garg & Anand, 2015). However, much of this work remains descriptive or focuses on supply-side interventions, such as product availability and distribution (Babbar et al., 2023). Far less attention has been paid to how women’s life-course trajectories shape menstrual health and hygiene (MHH) practices. In particular, the

role of major lifecycle transitions has been largely overlooked. This paper examines one such critical lifecycle factor: women’s age at marriage. We argue that the timing of marriage may be a fundamental determinant of period product usage, operating through multiple interconnected pathways.

Early marriage, which remains prevalent in India despite legal prohibitions, interrupts girls’ education and limits their exposure to health information and awareness about MHH (Field & Ambrus, 2008; Hicks & Hicks, 2019). It also constrains women’s decision-making power within households and restrict their control over resources (Roychowdhury & Dhamija, 2021), thereby reducing their ability to prioritize and access period products. On the other hand, delayed marriage may create a critical window for human capital accumulation, greater exposure to information through schooling and peers, and enhance bargaining power within households. These changes plausibly increase the likelihood of adopting MHH practices. Despite the intuitive appeal of these mechanisms, establishing a causal relationship between age at marriage on MHH practices is challenging. Women’s marriage timing is endogenous, shaped by household socioeconomic conditions, cultural norms, and individual characteristics that may themselves directly influence MHH practices.

To address this identification challenge, we adopt an instrumental variable (IV) approach that exploits exogenous variation in women’s age at menarche i.e., the onset of menstruation. This strategy, pioneered by (Field & Ambrus, 2008) and employed in subsequent studies on age at marriage (Chari et al., 2017; Dhamija & Roychowdhury, 2020; Roychowdhury & Dhamija, 2021; Sekhri & Debnath, 2014), relies on the fact that age at menarche is largely driven by genetic and biological factors beyond individual control, yet strongly predicts marriage timing in settings where parents seek to arrange daughters’ marriage soon after they reach reproductive maturity (Caldwell et al., 1983; Wahhaj, 2018). Using individual-level data from NFHS-5, we leverage this source of variation to estimate the causal effect of women’s age at marriage on their likelihood of using period products.

Our findings are noteworthy. We find that a one-year increase in a woman’s age at marriage leads to 7 percentage point increase in the likelihood of period product use. This result is robust to a wide set of controls, alternative constructions of the outcome variable, and falsification exercises. Sub-sample analyses reveal that the effects are particularly pronounced among women belonging to backward social groups and residing in rural areas. We also explore potential mechanisms and provide suggestive evidence that women’s age at marriage influences the likelihood of period product use by increase in women’s educational attainment and enhancing women’s decision-making power within the household.

This work contributes to two main strands of literature. First, it contributes to burgeoning literature on the determinants of MHH. Evidence from India highlights that MHH outcomes are shaped by intersecting socioeconomic, infrastructural, and cultural factors. NFHS-5 data shows schooling, autonomy, employment, media exposure, sanitation, asset index, urban residence, and regional location as the strongest predictors of adequate MHH (Goli et al., 2020; Meher & Sahoo, 2023; Roy et al., 2021; Singh et al., 2022; Zode et al., 2025). District-level access to MHH ranges from 2.3% to 89.4%, with a complementary spatial analysis documenting significant clustering of high- and low-access areas (Chakrabarty & Singh, 2024; Chakrabarty et al., 2025). Broader reviews also emphasize persistent cultural stigma, infrastructural deficits, and limited awareness as barriers to menstrual hygiene (Eijk et al., 2016; Sharma et al., 2020). Our study advances this literature by providing the first causal evidence on the effect of women’s age at marriage on their use of period products, drawing on large-scale nationally representative micro data.

Second, it adds to the growing body of research on the effects of age at marriage across a wide range of outcomes, including educational attainment (Field & Ambrus, 2008; Hicks & Hicks, 2019; Sekhri & Debnath, 2014), health outcomes (Ahmad & Vollmer, 2024; Babbar et al., 2025; Chari et al., 2017; Goli et al., 2015; Jayawardana, 2022; Maiti et al., 2024; Tafesse et al., 2025), nutritional outcomes (Efobi, 2024), labour market outcomes (Dhamija & Roychowdhury, 2020; Sunder, 2019), domestic violence (Roychowdhury & Dhamija, 2021), and women empowerment (Tauseef & Sufian, 2024). Given that child marriage remain pervasive concerns across many developing countries, examining their broader implications is vital for understanding women’s health and socioeconomic well-being. By focusing on the relationship between women’s age at marriage and their use of period products, this study extends current knowledge on the lesser-explored health and behavioural consequences of early marriage. Our findings offer policy-relevant insights for governments and development stakeholders seeking to evaluate and design interventions aimed at delaying marriage among young women, particularly in contexts where early marriage continues to pose significant barriers to women’s autonomy and health.

The remainder of the paper proceeds as follows. Section 2 describes the data and presents descriptive statistics. Section 3 outlines our empirical strategy. Section 4 presents the base-line results, while Section 5 reports robustness checks and alternative specifications. Section 6 explores the potential mechanisms. Section 7 examines heterogeneity across population sub-groups. Finally, Section 8 concludes with a discussion of policy implications and directions for future research.

2 Data

We use the fifth round of the National Family Health Survey (NFHS-5) conducted in India during 2019-21 (IIPS & ICF, 2021). NFHS-5 is a nationally representative, cross-sectional survey and forms part of the global Demographic and Health Survey (DHS) program.¹ In India, the survey is implemented by the International Institute for Population Sciences, Mumbai, under the supervision of the Ministry of Health and Family Welfare, Government of India. The sampling design follows a stratified random sampling.²

In addition to household questionnaire, a separate women’s questionnaire was administered to 724,115 eligible women aged 15-49 years residing in 636,669 households across 707 districts. The woman’s questionnaire included questions on a wide range of topics including demographic characteristics, fertility and family planning, nutrition, marriage, sexual activity, employment, mobility, autonomy, and menstrual hygiene practices. However, questions on menstrual hygiene were administered to 241,180 women aged 15-24.

2.1 Period Product Usage

Our primary outcome variable captures women’s choice of period product. In NFHS-5, respondents were asked: “*Women use different methods of protection during their menstrual period to prevent blood stains from becoming evident. What do you use for protection, if anything?*”. Women could select multiple options from six possible responses: *Cloth, Locally prepared napkins, Sanitary napkins, Tampons, Other, Nothing*. Figure 1 shows the distribution of products used by women in our sample. The most commonly reported products are sanitary napkins (57%), cloth (59%), or locally prepared napkins (14%).

Following the existing literature (Babbar et al., 2024; Biswas et al., 2024; Karjee et al., 2023), we classify women as *users* of period products if they report using *menstrual cups, tampons, locally produced napkins, or sanitary napkins*, and as *non-users* if they report exclusive use of *cloth or other objects or nothing*. Because multiple responses are allowed, our primary outcome is coded as one if a woman uses at least one period product and zero otherwise.

2.2 Treatment Variable and Control Variables

Our main explanatory variable of interest is women’s age at marriage (in years), denoted as *MarriageAge*. Figure 2 shows its distribution in our sample. To account for potential confounding factors that could influence the relationship between women’s age at marriage and their use

¹The DHS surveys for all countries are available at <https://dhsprogram.com/>.

²See IIPS and ICF (2021) for more details on the survey methodology.

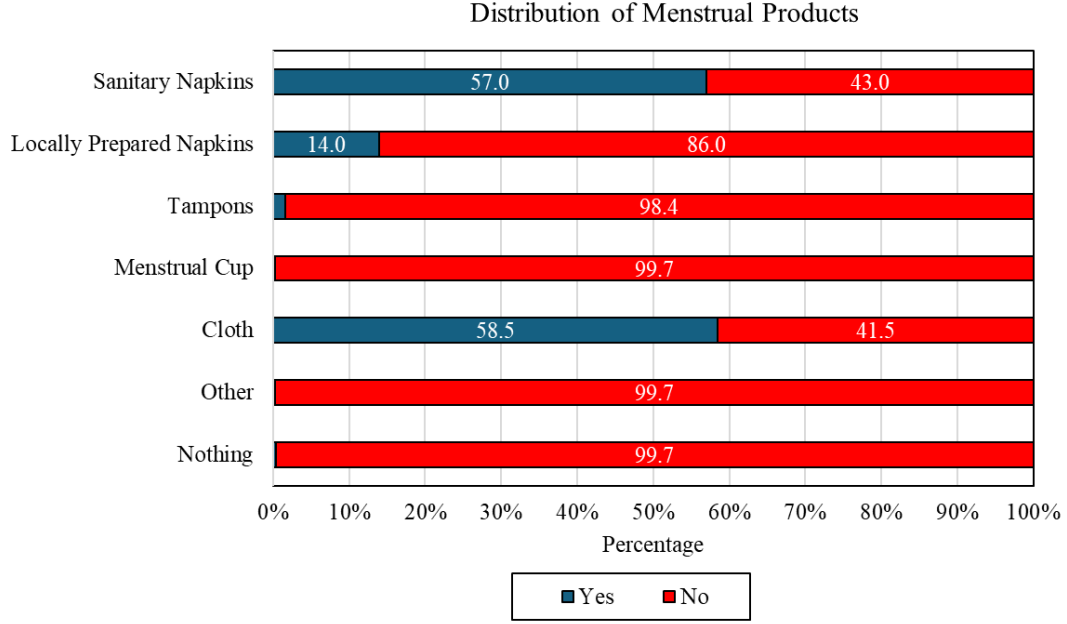


Figure 1: Distribution of various menstrual products used by women in India, 2019-21.

of period products, we include a range of individual, household, and cluster³-level geographic characteristics (Oster, 2019). Individual-level controls include woman’s age (in years), height (in cm), and exposure to mass-media (yes/no). Household-level controls include sex of household head (male/others), number of household members, religion (Hindu/Muslim/Others), social group (Scheduled caste/Scheduled tribe/Other Backward Class/others or none /don’t know or missing), wealth category (poor/non-poor)⁴, and area of residence (rural/urban). These controls are included based on existing literature on determinants of period product usage in India (Babbar et al., 2021; Biswas et al., 2024; Ram et al., 2020; Roy et al., 2021)

We further include cluster-level ecological and infrastructural characteristics that may jointly shape marriage timing and menstrual practices. These include elevation, terrain ruggedness, and nighttime light intensity. Elevation data is sourced from the Shuttle Radar Topography Mission (SRTM) at 1-arc second resolution. Ruggedness is measured as the elevation difference between two surfaces (Nunn & Puga, 2012; Riley et al., 1999). Nighttime lights, which serve as a proxy for local economic activity and urbanization (Dhamija et al., 2025), are measured in average cloud-free radiance values (Mayala & Donohue, 2022; Mayala et al., 2018). It comes from annual nighttime light intensity rasters recorded by the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band (DNB), operated jointly by National Aeronau-

³Clusters in NFHS are typically villages in rural areas or census enumeration blocks (CEBs) in urban areas, serving as the first-stage units from which households are sampled.

⁴The DHS wealth index is a standardized measure of household economic status that classifies households into five groups: poorest, poor, middle, richer, and richest. We group the poorest and poor categories as *poor* and the remaining three as *non-poor*.

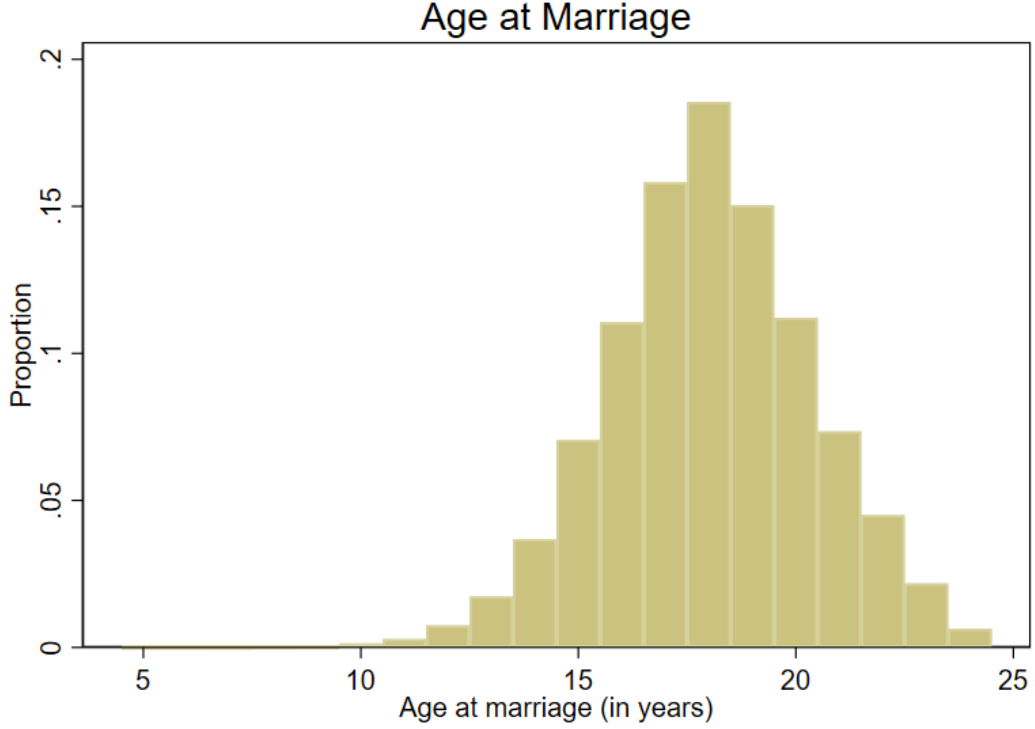


Figure 2: Distribution of age at marriage

tics and Space Administration (NASA) and National Oceanic and Atmospheric Administration (NOAA) (Elvidge et al., 2017). To mitigate potential simultaneity concerns, we use one-year lagged values of nighttime lights corresponding to the survey year for each cluster.

2.3 Analytical Sample

We use the individual recode file of NFHS-5, which contains one record for each eligible woman respondent. Since information on period products was collected only from women aged 15–24, we begin with a sample of 82,956 ever-married women in this age group. We then construct the analytical sample through a series of restrictions. First, we drop 120 observations with missing information on the outcome and an additional 623 observations with missing age at marriage. Second, following the existing literature (Dhamija & Roychowdhury, 2020; Roychowdhury & Dhamija, 2021), we exclude 623 women whose reported marriage age is below five, and age at menarche lies outside the biologically plausible range of 9–21 years. Third, adding women’s height as a control variable results in the loss of 2,314 observations, and including cluster-level controls eliminates an additional 1,120 observations. The final analytical sample includes 78,779 women. A schematic description of sample construction is provided in Figure A1.

2.4 Summary statistics - Analytical sample

Table 1 presents summary statistics for our analytical sample. On average, 70% of women use at least one period product. The average respondent is 21.6 years old, married at an age of 18 years, and around 73% of women report some exposure to mass media. The average height of women in our sample is around 151.6 cm. Approximately, 85% of the households are male-headed, with an average household size of six members. The religious composition shows that 79% of households are Hindu, 13% Muslim, and 8% belong to other religions. The social group composition is diverse, with 22% are Scheduled Castes (SC), 18% are Scheduled Tribes (ST), 41% are Other Backward Classes (OBC), and 14% belong to other categories. Around 52% fall in the poorest and poor categories of the wealth index and 84% reside in rural areas. Cluster-level characteristics show that the mean elevation is 349.74 m, terrain ruggedness averages 7.54 m, and average nightlight intensity is 1.68 units.

3 Empirical framework

3.1 Ordinary least squares estimation

We begin with the following specification to investigate the effect of women’s age at marriage on their likelihood of using period products:

$$PPU_i = \beta_0 + \beta_1 \text{MarriageAge}_i + \beta_2 \mathbb{X}_{ih} + \beta_3 \mathbb{G}_c + \theta_d + \epsilon_{ihcd} \quad (1)$$

where PPU_{ihdst} captures the period product use of woman i ; MarriageAge_i denotes the woman’s age at marriage; $\mathbb{X}_{ih}$ is a vector of individual and household characteristics; $\mathbb{G}_c$ is a vector of cluster-level geographic characteristics; θ_d are district fixed effects to account for any systematic district-level time-invariant unobservables potentially correlated with age at marriage and period product use. ϵ_{ihcd} is the idiosyncratic error term. Standard errors are clustered at the cluster level.

Our parameter of interest, β_1 , captures the association between age at marriage and period product use. A causal interpretation of β_1 requires that no unobserved variables are simultaneously correlated with both age at marriage and period product usage, conditional on the included controls. If such endogeneity is present, ordinary least squares (OLS) estimate of β_1 would be biased and inconsistent. Our empirical specification incorporates several layers of controls to address potential confounders. First, we include an extensive set of woman- and household-level characteristics. Second, cluster-level geographic and infrastructural char-

acteristics control for ecological and infrastructural conditions, such as terrain and ruggedness, that may directly influence the period product usage. Third, district fixed effects account for time-invariant district-level factors, such as cultural norms, that simultaneously shape age at marriage and MHH practices.

Despite these controls, the possibility of omitted variable bias cannot be fully ruled out. For instance, women from patriarchal households may marry younger and have limited autonomy in choosing period products. Similarly, women’s innate ability might affect both marriage timing and MHH practices. Additionally, age at marriage is self-reported. Any misreporting introduces measurement error in the explanatory variable. This misreporting could bias the estimated OLS coefficient toward zero. Thus, we employ an instrumental variables approach to estimate the causal effect of age at marriage on period product usage.

We instrument women’s age at marriage with their age at menarche. Menarcheal timing triggers an anxiety response from parents (Caldwell et al., 1983; Ortner, 1978; Wahhaj, 2018). Families worry about premarital sexual activity or the risk of an unwanted pregnancy once a daughter attains menarche. These concerns are tied to fears of damaged family honour, and trigger pressure to arrange marriage soon after menarche. Moreover, Field and Ambrus (2008) argues that age at menarche can be treated as exogenous because much of its variation reflects underlying genetic differences rather than parental choices or social factors (Kaprio et al., 1995). A rich body of literature has used age at menarche as an instrument for age at marriage (Babbar et al., 2025; Chari et al., 2017; Dhamija & Roychowdhury, 2020; Field & Ambrus, 2008; Roychowdhury & Dhamija, 2021; Sekhri & Debnath, 2014). Table 1 shows that average age of menarche in our analytical sample is 13.5 years. Figure 3 presents the distribution of age at menarche in our sample.

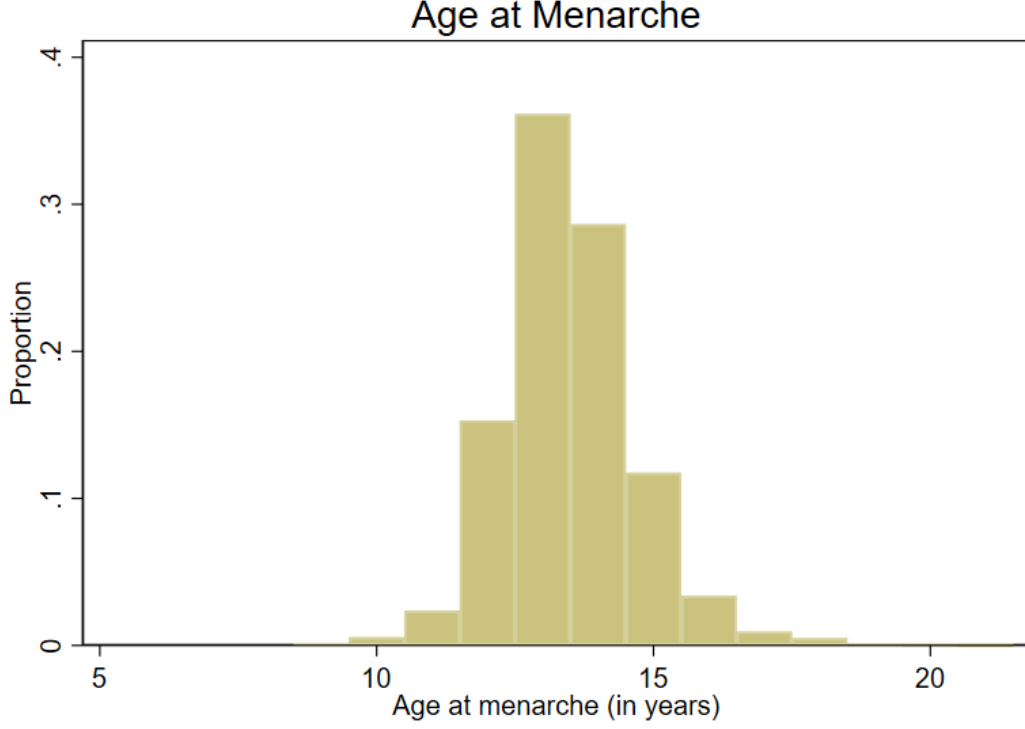


Figure 3: Distribution of age at menarche

3.2 Instrumental variable estimation

We estimate a two-stage IV model, which is specified as follows:

$$MarriageAge_i = \alpha_0 + \alpha_1 MenarcheAge_i + \alpha_2 \mathbb{X}_{ih} + \alpha_3 \mathbb{G}_c + \phi_d + \eta_{ihcd} \quad (2)$$

$$PPU_i = \beta_0 + \beta_1 MarriageAge_i + \beta_2 \mathbb{X}_{ih} + \beta_3 \mathbb{G}_c + \theta_d + \epsilon_{ihcd} \quad (3)$$

The first stage for the IV regression is given by Equation 2, and Equation 3 is the structural equation. A woman's age at marriage, $MarriageAge_i$, is instrumented by $MenarcheAge_i$, woman's age at menarche. As above, $\mathbb{X}_{ih}$ and $\mathbb{G}_c$ are the vectors of woman/household level characteristics and cluster-level geographic characteristics, respectively. ϕ_d or θ_d denote district fixed effects. We use two-stage least squares (TSLS) estimation and cluster standard errors at the cluster level. Our parameter of interest, β_1 , captures the local average treatment effect (LATE) of women's age at marriage on period product usage. Specifically, it identifies the effect for women whose age at marriage varies as a consequence of age at menarche.

3.3 Validity of the IV

For age at menarche to serve as a valid instrument for age at marriage, our IV must satisfy three conditions: *relevance*, *exclusion restriction*, and *monotonicity*. First, it must be strongly correlated with age at marriage. Second, conditional on controls, it must influence period product use only through its effect on marriage timing. Third, the relationship between age at menarche and age at marriage is always in the same direction (Huntington-Klein, 2021).

To examine the validity of the instrument, we begin by testing whether age at menarche is a strong predictor of women’s age at marriage. Figure 4 plots mean age at marriage against mean age at menarche with a fitted linear trend. The upward slope indicates a clear positive association. Figure 5 further illustrates this relationship by comparing kernel density estimates of age at marriage for women experiencing early versus late menarche, defined relative to the sample median age of menarche. The early-menarche group includes women who attained menarche at age 13 or earlier, while the late-menarche group consists of those who did so after age 13. Women with later menarche exhibit a rightward shift in the marriage-age distribution, consistent with delayed marriage. Together, these figures provide preliminary evidence of instrument relevance.

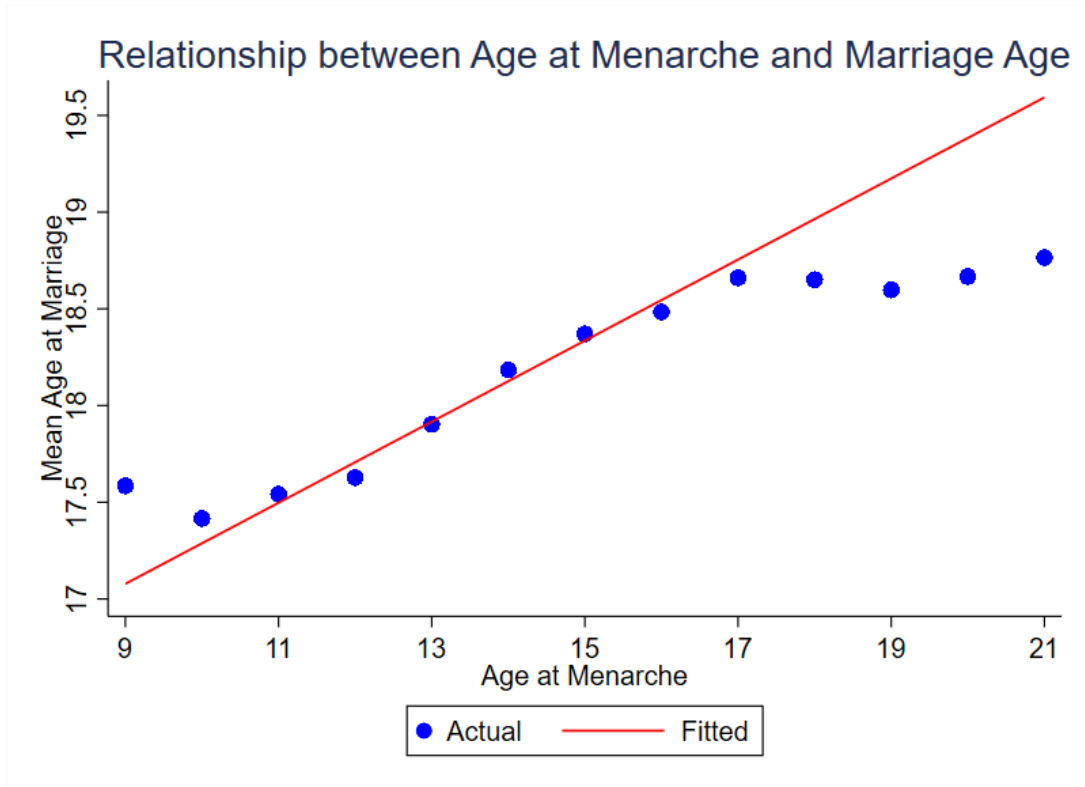


Figure 4: Correlation between age at marriage and age at menarche

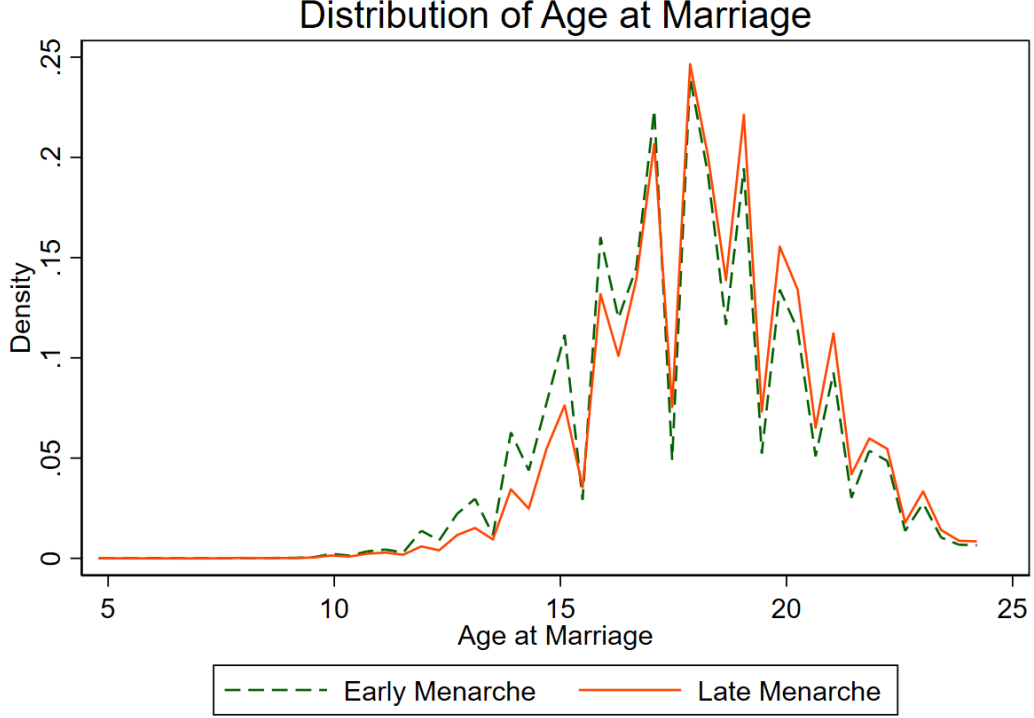


Figure 5: Distribution of women’s age at marriage by age at menarche

Table 2 presents first-stage results based on Equation 2. In the unadjusted specification (Column 1), we find a strong positive correlation between age at menarche and age at marriage (coefficient=0.25, p -value < 0.01) with an F-statistic of 673.21. The association continues to be positive and significant even after sequential addition of individual/household characteristics, cluster-level characteristics, and district fixed effects (columns 2–4). In the final specification (column 4), a one-year increase in age at menarche is associated with roughly a 0.14-year delay in age at marriage (p -value < 0.01). The corresponding F-statistic (255.59) and Kleibergen–Paap rk LM statistic (241.09) firmly reject weak and under-identification of our IV. These results establish the relevance of age at menarche as an instrument for age at marriage.

We also argue that age at menarche does not directly influence period product use. A potential concern is that women from disadvantaged social groups may experience delayed menarche due to childhood malnutrition (Nandi et al., 2020) or physically demanding labour (Pellerin-Massicotte et al., 1987), while also being less likely to use period products because of affordability or access constraints. Additionally, geographic and climatic factors may also influence menarcheal timing (Canelón & Boland, 2020; Dossus et al., 2013; Field & Ambrus, 2008; Nishan et al., 2025) while simultaneously shaping infrastructure and market access for period products and services (Maithreyi, 2025). These factors can lead to potential violations of the exclusion restriction if they are not properly accounted for. To minimize the potential

endogeneity of menarche, we include height as a proxy for early-life nutritional status, social group and husband's household wealth as proxies for natal socioeconomic conditions in the natal household (Chari et al., 2017; Roychowdhury & Dhamija, 2021). Given emerging evidence of an association between media exposure and timing of menarche (Fuadah, 2020; Novianti & Ardila, 2021), we additionally control for women's exposure to mass media. To account for spatial variation in ecological conditions that influence menarcheal timing, we further include cluster-level measures of ruggedness, elevation, along with district fixed effects. Conditioning on this rich set of controls substantially reduces the scope for direct pathways linking menarche to menstrual hygiene practices.

A second concern is that earlier menarche may mechanically increase exposure to period products. Thus women with earlier menarche are more likely to adopt such products irrespective of their age at marriage. To assess whether this threatens our identification strategy, we conduct a residual-based test. First, we regress period product use on age at marriage and extract the residuals. Second, we regress age at menarche on age at marriage and obtain the corresponding residuals. Figure 6 plots these two sets of residuals. The absence of any systematic pattern suggests that, conditional on age at marriage, age at menarche is not directly associated with period product use.

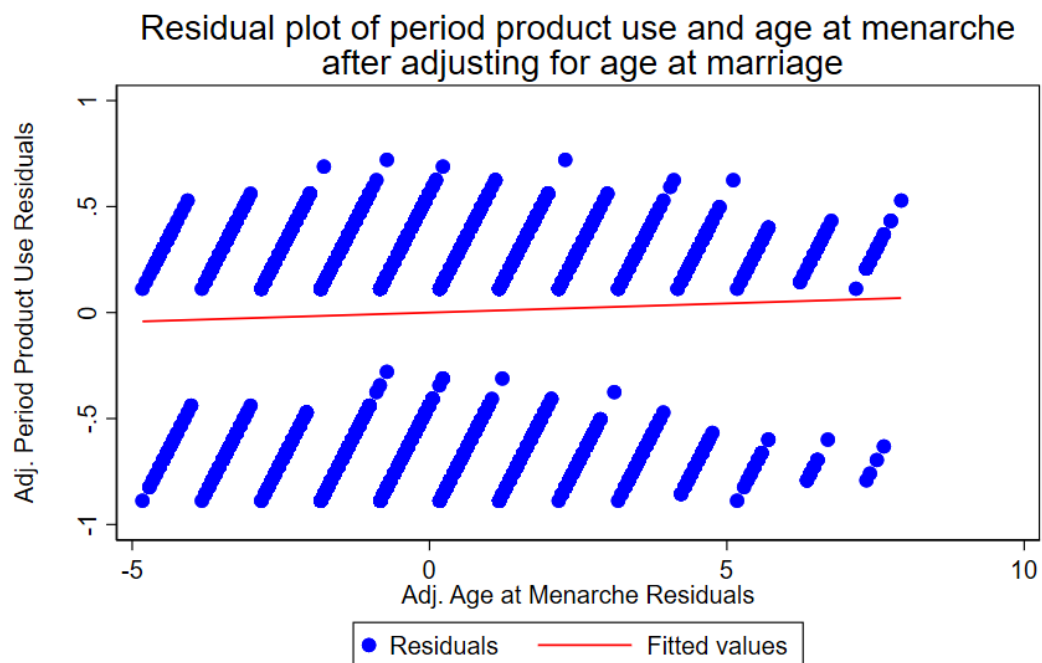


Figure 6: Residual Plot: Use of period products vs. Age at Menarche (Adjusted for Age at Marriage)

Finally, we examine the validity of monotonicity condition by estimating the first-stage

relationship across multiple subsamples defined by social group (Backward/Others), women’s age (older/younger)⁵, place of residence (urban/rural),and wealth status (non-poor/poor). The corresponding coefficients along with the 95% confidence intervals are presented in Figure 7. Across all subsamples, the association between age at menarche and age at marriage remains consistently positive and statistically significant. This uniformity across heterogeneous subpopulations provides strong support that the instrument satisfies the monotonicity assumption.

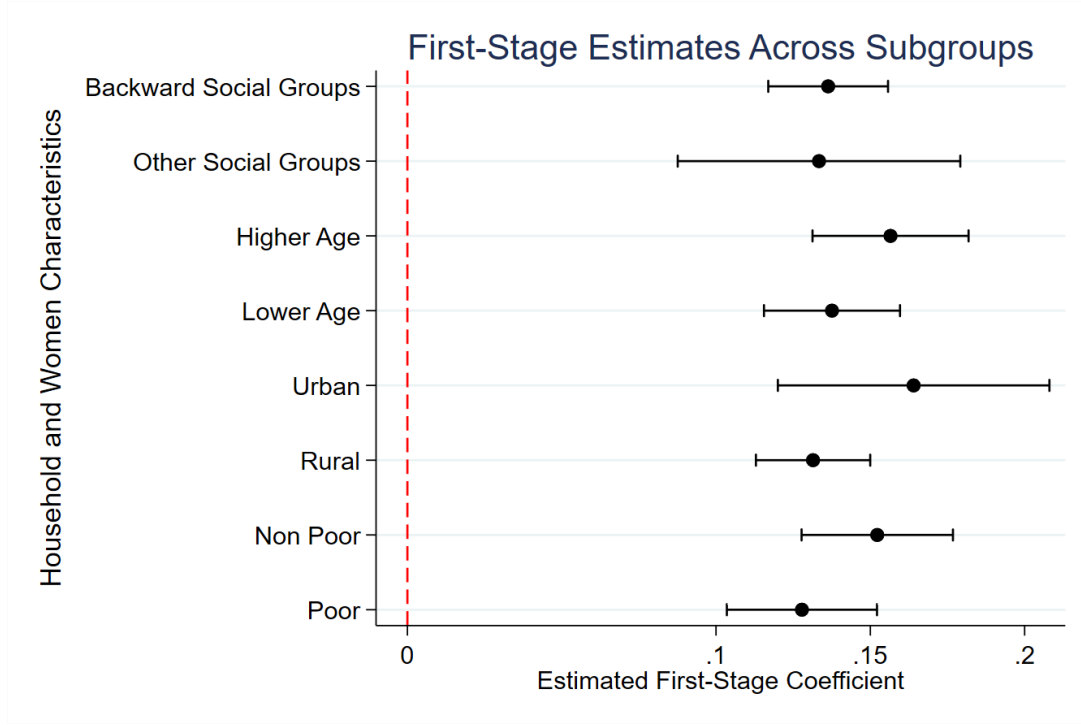


Figure 7: First-stage relationship across multiple subsamples

4 Results

4.1 Baseline results

Table 3 reports the baseline estimates. Panel A (columns 1–4) presents the OLS results, while Panel B (columns 5–8) presents the IV estimates. In the naïve specification (Column 1), where period product use is regressed solely on age at marriage, we find a positive association. The estimated coefficient remains stable as individual, household, and cluster-level controls are added sequentially (columns 2–3), and when district fixed effects are also included (column 4). In the fully adjusted specification (column 4), a one-year increase in age at marriage is associated with a 2 percentage point (pp) higher likelihood of using period products. Overall, the OLS results consistently indicate a positive association between age at marriage and period

⁵Women younger than 22 years (the sample median) are classified as the younger cohort, while those aged 22 and above form the older cohort.

product use.

In panel B, we present the IV estimates following the same progression as in panel A. Column (5) presents the naïve specification, and column (6) includes individual and household-level controls. Columns (7) and (8) further incorporate cluster-level characteristics and district fixed effects, respectively. Across all IV specifications, age at marriage has a positive and statistically significant effect on period product use. Our preferred specification (column 8) shows that a one-year delay in age at marriage raises the likelihood of using period product by 7 pp. Substantively, this estimate implies that increasing women’s age at marriage by one year nationwide would raise period product use from 70% to roughly 77%. This represents a meaningful improvement in menstrual health practices among young women in India.

5 Potential mechanisms

In this section, we examine plausible mechanisms through which age at marriage affects period product use. We focus on two channels that are biologically and socially salient over the female life course: educational attainment and intra-household decision-making power. Delaying marriage extends the period during which girls can remain in school (Field & Ambrus, 2008; Hicks & Hicks, 2019; Roychowdhury & Dhamija, 2021). Additional schooling may deepen menstrual hygiene knowledge and awareness of available products, thereby increasing the likelihood of adopting MHH practices.

To empirically test education as a channel, we estimate an IV model to capture the effect of women’s age at marriage on their years of education. Column (1) of Table 6 shows that a one-year increase in age at marriage increases years of education by 1.21 years. This large effect reflects that in our sample, marriage timing and educational attainment are closely intertwined, early marriage often coincides with school dropout. Following Bose et al. (2020) and Mookerjee et al. (2023), we then use the predicted values of schooling in a linear regression of period product use. We bootstrap the standard errors with 100 replications to account for the generated regressor. Column (2) shows that predicted education is positively and significantly associated with the likelihood of using period products.

We next examine whether women’s household decision-making power operates as a second mechanism. NFHS-5 records whether women have sole, joint, or no say over decisions related to (a) their own healthcare, (b) major purchases, (c) visits to family or relatives, and (d) use of the husband’s earnings. For each indicator, we code sole or joint decision-making as one and no say as zero. Subsequently, we use these components to construct a standardized inverse-probability-weighted (IPW) index of decision-making power. Delay in marriage is known to

improve women’s bargaining power and reduce gender asymmetries within households (Tauseef & Sufian, 2024). Greater autonomy may, in turn, enhance women’s ability to choose use period products.

Column (3) of Table 6 reports the IV estimate of the effect of age at marriage on this decision-making index. A one-year delay in marriage increases the decision-making power by 0.16 standard deviations. Using the fitted values of this index in a subsequent regression of period product use with 100 bootstrap replications, we find that greater decision-making power increases the likelihood of using period products by 12 percentage points (column 4).

6 Robustness

6.1 Disaggregated Outcome Variable

As noted earlier, we classify sanitary napkins, locally produced napkins, tampons, and menstrual cups as period products. In this section, we examine whether the effect of age at marriage varies across specific types of period products. For each outcome, we construct a binary indicator equal to 1 if a woman reports using that period product and 0 if she reports using only non period product alternatives such as cloth, other, or nothing. Table 4 presents the IV estimates for sanitary napkins, locally produced napkins, tampons, and menstrual cups in columns (1) to (4), respectively. A one-year increase in age at marriage raises the probability of using sanitary napkins by 7 pp and locally produced napkins by 10 pp. The effect on menstrual cup use is positive but small (1 percentage point), reflecting the very low baseline adoption of cups in the analytical sample. We find no meaningful effect on tampon use. These results indicate that delayed marriage primarily increases uptake of products that are relatively affordable, widely available, and culturally familiar, rather than those requiring greater informational or behavioural change.

6.2 Alternative Outcome variable

Our baseline outcome classifies women as users if they report using at least one of the following period product: menstrual cups, tampons, locally produced napkins, or sanitary napkins. Based on this construction, women who report using cloth or other objects along with menstrual cups, tampons, locally produced napkins, or sanitary napkins are also classified as users. To assess sensitivity to this choice, we construct two additional measures of period product use. First, we define the outcome to equal one only if a woman reports *exclusive* use of menstrual cups, tampons, locally produced napkins, or sanitary napkins, and zero if she reports using

cloth, other objects or no period product. In comparison to the baseline construction, women who report mixed use (using cloth or other objects along with menstrual cups, tampons, locally produced napkins, or sanitary napkins) are classified as non-users. Second, we adopt a more *restrictive* definition by excluding women who report using cloth or other objects along with menstrual cups, tampons, locally produced napkins, or sanitary napkins from the sample. In this case, women who use only menstrual cups, tampons, locally produced napkins, or sanitary napkins are coded as one, while those who use cloth, other objects or no period product are coded as zero.

Table 5 reports the corresponding IV estimates in columns (1) and (2). Under the exclusive-use definition (column 1), a one-year increase in age at marriage raises the probability of *exclusive* use of period product by 13 pp. Under the restricted-sample definition (column 2), a one-year increase in age at marriage increases the probability of period product use by 11 pp. Across both alternative definitions, the estimated effects are larger than those obtained under the baseline specification. This pattern suggests that delaying marriage has a stronger impact on the consistent adoption of period products.

6.3 Alternative Estimation Strategy

Our baseline results rely on the IV TSLS approach. Given the binary nature of the outcome variable, we additionally estimate a non-linear specification to assess whether the results are sensitive to our estimation strategy. Specifically, we use a Probit model with endogeneity addressed through a control function approach proposed by Rivers and Vuong (1988) and later developed by Blundell and Powell (2004) and Wooldridge (2010, 2015). We continue to use the same set of controls as in the baseline. Column (3) of Table 5 reports the corresponding average marginal effect. It shows that a one-year increase in age at marriage raises the probability of using period products by about 8 pp. The magnitude is in-line with our TSLS estimate, providing further support for the robustness of our results.

6.4 Trimming Analytical Sample

We further assess the sensitivity of our baseline estimates by trimming both the treatment variable and the instrumental variable to mitigate the influence of outliers. Specifically, we sequentially exclude the top and bottom 1%, 5%, and 10% of the distribution of women’s age at marriage (columns 1-3 of Table 7) and, separately, of age at menarche (columns 4-6 of Table 7). Across all trimming thresholds, the estimated effect of age at marriage on period product use remains positive and statistically meaningful. When trimming age at marriage, the implied

increase in product use due to one-year delay in marriage ranges from 8 pp to 19 pp. Similarly, trimming the instrument yields stable estimates, with effect size ranges between 8 and 10 pp. Overall, these results indicate that neither extreme values in age at marriage nor in age at menarche drive our main findings.

Next, we examine whether our results are sensitive to disruptions associated with the COVID-19 pandemic. The pandemic substantially disrupted MHH practices, with reports indicating that many girls and women substituted away from sanitary products as prices rose and access deteriorated (Xiao & Darmadi, 2020). Evidence from Menstrual Health Alliance India documents widespread supply-side disruptions during this period, including production shutdowns, diversion of manufacturing capacity toward masks, retail closures, hoarding, black-market activity, and sharp price increases, which constrained access to sanitary products in both rural and urban areas even after restrictions were relaxed (Chander, 2021; Sanand, 2020). Consistent with these disruptions, Babbar et al. (2023) show that stricter lockdowns led to large declines in sanitary pad distribution to adolescent girls. Thus, we estimate our results for two subsamples in columns (7) and (8) of Table 7: a) observations collected prior to the onset of pandemic-related restrictions (till March 2020) and b) after the onset of pandemic-related restrictions (from April 2020) respectively. Reassuringly, both the columns show that our estimates are virtually identical to our baseline results.

6.5 Additional controls

Next, we evaluate the robustness of our baseline estimates by introducing additional controls. First, we account for husband’s characteristics which may be correlated with both women’s age at marriage and their period products usage. Specifically we include husband’s current age, education and employment status. Such factors have been shown in prior literature to shape women’s bargaining power and gender norms within households (Mahon et al., 2015; Pourtaheri et al., 2023). As shown in column (1) of Table 8, the effect of women’s age at marriage on the use of period products remains positive and statistically significant after adjusting for these controls. The magnitude is close to the baseline estimate, with a one-year increase in age at marriage raises the probability of using period products by 8 pp.

Second, we include a variable indicating whether the respondent woman witnessed her father physically abuse her mother. Childhood exposure to parental violence may affect and the timing of menarche (Henrichs et al., 2014). Moreover, it is also a proxy for the socioeconomic status of the woman’s natal home, which in turn can affect the woman’s age at menarche and her use of period products. Column (2) of Table 8 shows that accounting for this factor does not

materially alter the estimated effect of age at marriage on period product use.

6.6 Falsification test

We conduct two sets of randomization tests to address concerns about spurious correlation and to further validate our identification strategy. Our baseline results consistently show a positive and statistically significant effect of women’s age at marriage on their likelihood of using period products. In the first test, we randomly permute women’s age at marriage across individuals. This procedure breaks the first-stage relationship by disrupting the link between age at marriage and age at menarche. It also breaks the structural equation relation between marriage timing and the likelihood of using period products.

In the second test, we randomize the outcome variable by shuffling the binary indicator for period product usage across women. This preserves the first stage relationship while breaking the structural relationship between age at marriage and period product usage. In both cases of falsification, the randomization deliberately destroys the pathways through which a causal effect could plausibly operate, thereby serving as stringent falsification tests.

We replicate each randomization test 100 times and plot the resulting distribution of t-statistics in Fig 8. In Panel A, we do not find any statistically significant impact of age at marriage on period product usage in all of the 100 iterations. Panel B shows the results from randomizing the outcome variable. Here, we fail to reject the null effect of age at marriage in 6 iterations. In 45% of iterations, we do not find a statistically significant effect of age at marriage on period product usage when the outcome variable is randomized. Together, these results indicate that significant effects almost never arise when the underlying causal structure is deliberately broken. These results strongly suggest that our main findings are not artefacts of random chance and provide further support for the credibility of our identification strategy.

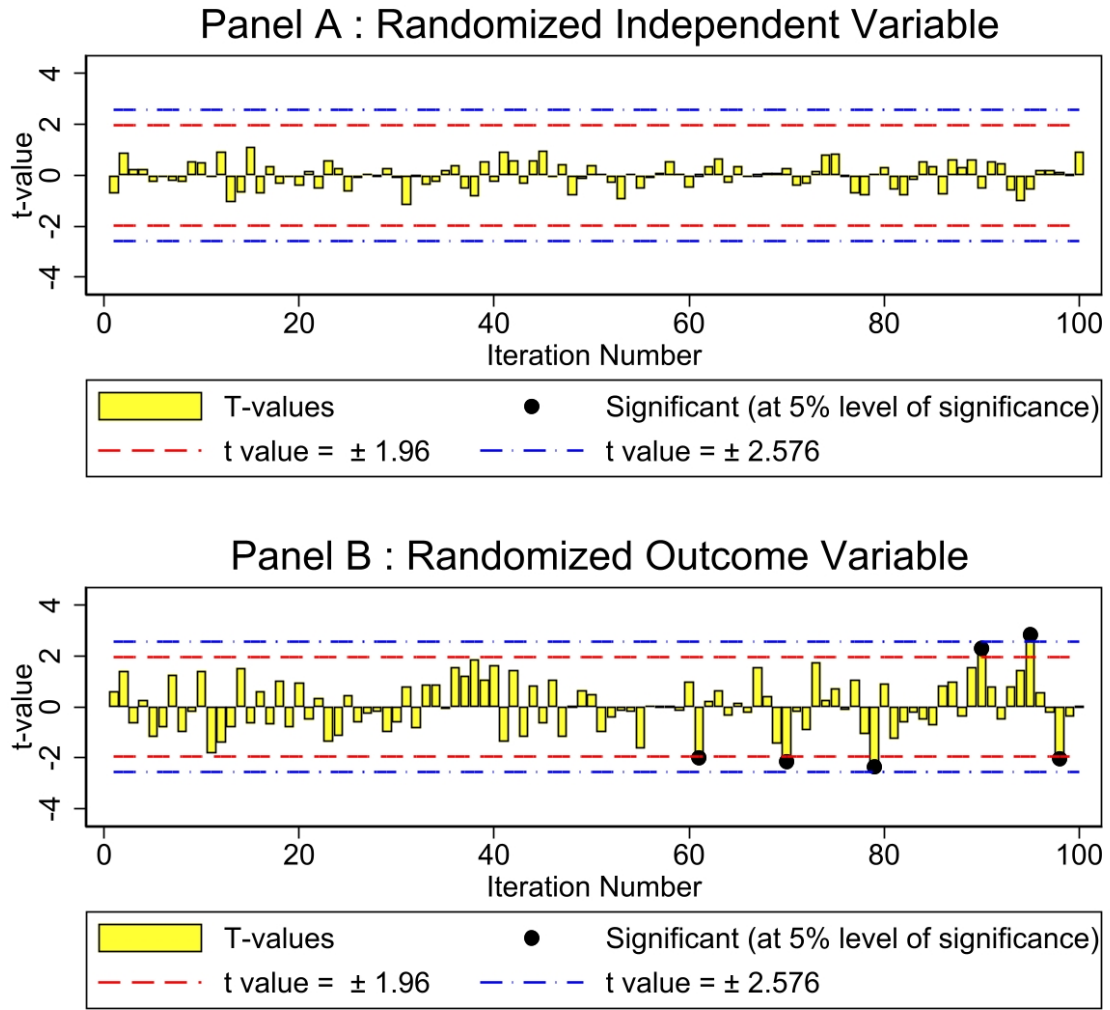


Figure 8: Falsification Analysis

7 Sub-sample analysis

Last, we examine whether the effect of age at marriage on period product use varies across key sub-populations by cutting our analytical sample in several ways. Specifically, we conduct sub-sample analysis by women's age, household wealth index, social group, and area of residence. For each subgroup, we re-estimate the IV specification using the full set of covariates and district fixed effects.

Figure 9 reports heterogeneity by these individual- and household-level characteristics. We begin by estimating the coefficients for different age cohorts. Women younger than 22 years (the sample median) are classified as the younger cohort, while those aged 22 and above form the older cohort. The effect of age at marriage remains positive for both groups, with an estimated coefficient of 8.1 pp among younger women and 5.8 pp among older women. Heterogeneity by

household wealth indicates a positive effect of age at marriage across both poor and non-poor households.

We then present heterogeneity by social group. Given the well-documented role of social group on women’s autonomy and resource access (Deshpande, 2011), and the compounded disadvantages faced by marginalized groups, we classify women into backward social groups (SC/ST/OBC) and other social groups. We find that age at marriage has a statistically significant and positive effect on period product usage only among women belonging to backward social groups. It suggests that the effect is pronounced among women from historically disadvantaged backgrounds. Previous research indicates that the use of period products is significantly lower among women from historically marginalized and socially disadvantaged groups in India (Babbar et al., 2024). Women belonging to these communities are often identified as more vulnerable than those from other social categories (Sabharwal & Sonalkar, 2015). Moreover, individuals from these groups continue to lag behind in socioeconomic development and health outcomes, including healthcare access and utilization, with limited awareness frequently cited as a key contributing factor (Barman, 2020; Krishnan, 2005; Sankaran et al., 2017).

Finally, we estimate the effect separately for rural and urban samples. Rural and urban settings differ substantially in terms of contextual and structural factors that shape women’s access to period products. Across both residence types, the coefficient remains positive. However, the coefficient is imprecise for residing in urban areas. This rural–urban contrast is likely driven by differences in baseline adoption rates. Nearly 90% of urban women report using period products, compared to approximately 73% in rural areas. In urban settings, where access and awareness are already high, there is limited scope for delayed marriage to exert an additional effect. In contrast, affordability, distribution, and informational barriers remain substantial in rural areas (Priyadarshi & Ansh, 2025). Hence, marriage timing plays a more pronounced role in rural setting. Therefore we can observe that confidence intervals overlap across some subgroups, the pattern of coefficients consistently suggests larger effects among disadvantaged populations.

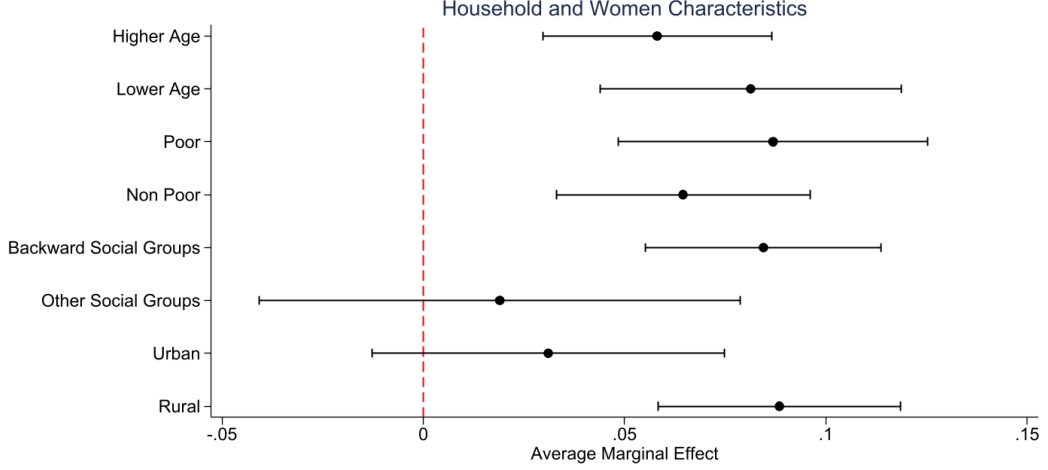


Figure 9: Sub-sample Analysis

8 Conclusion and Discussion

This paper provides causal evidence on the relationship between women’s age at marriage and menstrual health outcomes in India. Using nationally representative data from NFHS-5 (2019-21) and exploiting exogenous variation in age at menarche as an instrument for marriage timing, we find that delaying marriage by one year increases the probability of using period products by 7 pp. This finding is robust to alternative outcome specifications, estimation strategies, sample restrictions, and falsification tests.

We identify two key mechanisms through which age at marriage affects period product use. First, delaying marriage extends educational attainment, which increases knowledge and awareness of menstrual health practices. Second, women who marry later exhibit greater decision-making power within households, enhancing their ability to adopt health-promoting behaviors. Heterogeneity analysis reveals that effects are concentrated among socioeconomically disadvantaged groups, particularly women from backward castes and rural areas, where baseline adoption rates are lowest and constraints on autonomy are strongest.

The magnitude of our estimates has significant policy significance. A 7 pp increase in period product use represents a meaningful public health gain. To contextualize this effect, we note that among women aged 15–24 in India (approximately 18.1 million as per Ministry of Statistics and Programme Implementation, Government of India (2022)), roughly 55.88% are married (Office of the Registrar General & Census Commissioner, India, 2021). If marriage were delayed by one year nationwide, our estimates imply an increase of approximately 5 lakh additional users of period products among the population of this age group. While such extrapolation requires caution given potential general equilibrium effects and heterogeneity beyond what we observe,

it signifies the potential scale of impact.

Our findings have important implications for policy design. They show that interventions aimed at delaying child marriage can yield meaningful co-benefits for menstrual health, particularly among disadvantaged populations. Strengthening enforcement of the Prohibition of Child Marriage Act (2006) and expanding awareness campaigns may therefore generate downstream gains in health outcomes beyond the well-documented benefits for education and reductions in domestic violence. Our results lend support to conditional cash transfer programs that incentivize delayed marriage by keeping girls in school, such as Kanyashree Prakalpa (West Bengal) and Ladli Lakshmi Yojana (Madhya Pradesh). These programs address both mechanisms we identify, extending education and enhancing autonomy, making them particularly promising models for replication and expansion. However, we caution that legal changes alone, such as recent proposals to raise the minimum legal age of marriage to 21, may be insufficient without complementary investments in girls' education and efforts to challenge entrenched gender norms that drive early marriage. Our mechanism analysis suggests that the benefits of delayed marriage operate primarily through human capital accumulation and enhanced autonomy, implying that policies must address the root causes of early marriage rather than simply imposing age restrictions.

Our study has several limitations that present avenues for future research. First, NFHS-5 collects menstrual health data only for women aged 15-24, limiting the generalizability of our findings to all women of reproductive age. Second, while our findings are specific to India, the underlying mechanisms linking marriage timing to education, autonomy, and health behaviors are likely relevant in other settings where child marriage remains prevalent. Replicating this analysis in different cultural and institutional contexts would test the external validity of our results and inform whether similar policy interventions would be effective elsewhere. Finally, our analysis provides suggestive evidence on mechanisms but cannot fully trace the pathways through which delayed marriage translates into improved menstrual health practices. Despite these limitations, our findings make clear that marriage timing is a fundamental determinant of women's menstrual health outcomes, and interventions that delay child marriage represent a promising avenue for improving menstrual health and hygiene, particularly among disadvantaged populations.

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Table 1: Descriptive Statistics

Variable	Mean	SD	Min	Max
Outcome Variables				
Period Product Use				
Yes	0.70	0.46	0.00	1.00
No	0.30	0.46	0.00	1.00
Treatment Variable				
Age at Marriage	18.02	2.36	5.00	24.00
Women’s Characteristics				
Age (in years)	21.57	1.98	15.00	24.00
Height (in cm)	151.55	6.26	83.50	205.00
Media Exposure				
Yes	0.73	0.44	0.00	1.00
No	0.27	0.44	0.00	1.00
Household Characteristics				
Male-headed household				
Yes	0.85	0.36	0.00	1.00
No	0.15	0.36	0.00	1.00
Number of household members	5.88	2.65	1.00	31.00
Religion				
Hindu	0.79	0.41	0.00	1.00
Muslim	0.13	0.34	0.00	1.00
Other	0.08	0.27	0.00	1.00
Social Group				
Scheduled Caste	0.22	0.41	0.00	1.00
Scheduled Tribe	0.18	0.38	0.00	1.00
OBC	0.41	0.49	0.00	1.00
Other/None	0.14	0.35	0.00	1.00
Don’t Know/Missing	0.06	0.24	0.00	1.00
Wealth Category				
Poor	0.52	0.50	0.00	1.00
Non Poor	0.48	0.50	0.00	1.00
Residence				
Rural	0.84	0.37	0.00	1.00
Urban	0.16	0.37	0.00	1.00
Instrument Variable				
Age at Menarche	13.48	1.24	9.00	21.00
Observations	78,779			
Cluster Level Characteristics				
Ruggedness	7.54	9.04	1.53	61.23
Elevation	349.74	431.88	0.94	5188.15
Nightlights	1.68	3.32	0.00	46.76
Observations	24801			

Notes: Women are considered exposed to mass media if they report exposure to any of the following: newspaper, radio, or television. Other religions include Christian, Sikh, Buddhist/Neo-Buddhist, Jain, Jewish, Parsi/Zoroastrian, no religion, and others. The bottom two quantiles of the DHS wealth index (poorest and poor) are coded as poor, while the top two quantiles (richer, and richest) are coded as rich.

Table 2: First stage analysis : Regression of Age at marriage on Age at Menarche

	Age at Marriage			
	(1)	(2)	(3)	(4)
Age at Menarche	0.25*** (0.01)	0.17*** (0.01)	0.17*** (0.01)	0.14*** (0.01)
Observations	78,779	78,779	78,779	78,779
Woman's Characteristics	No	Yes	Yes	Yes
Household Characteristics	No	Yes	Yes	Yes
Geographical Characteristics	No	No	Yes	Yes
District FE	No	No	No	Yes
Kleibergen-Paap rK-LM Statistic	591.90	351.77	352.84	241.09
First Stage F Statistic	673.21	382.47	384.13	255.59
R-squared	0.02	0.16	0.16	0.21

Notes: Estimation via OLS analysis. Column (1) includes no additional controls. Column (2) adds women's and household characteristics. Column (3) additionally includes geographical characteristics, while Column (4) further includes district fixed effects. Women's characteristics include age, height and media exposure. Household characteristics include indicators for religion, social group, wealth category, age of the household head, household size, and area of residence. Geographic characteristics include ruggedness, elevation, night lights. Standard errors (in parentheses) are clustered at the neighbourhood level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 3: OLS and IV analysis: Estimates of the effect of Age at Marriage on Period Product Use

	Period Product Use							
	Panel A: OLS				Panel B: IV-TSLS			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Age at Marriage	0.03*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.05*** (0.01)	0.03** (0.01)	0.03*** (0.01)	0.07*** (0.01)
Observations	78,779	78,779	78,779	78,779	78,779	78,779	78,779	78,779
Woman's Characteristics	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household Characteristics	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographical Characteristics	No	No	Yes	Yes	No	No	Yes	Yes
District FE	No	No	No	Yes	No	No	No	Yes
Kleibergen-Paap rK-LM Statistic					591.90	351.77	352.84	241.09
First Stage F Statistic					673.21	382.47	384.13	255.59

Notes: Columns (1) - (4) presents the findings from the OLS estimation, whereas Columns (5) - (8) reports the results from the IV-TSLS estimation.. Column (1) and (5) includes no additional controls. Column (2) and (6) adds women's and household characteristics. Column (3) and (7) additionally includes geographical characteristics, while Column (4) and (8) further includes district fixed effects. Women's characteristics include age, height and media exposure. Household characteristics include indicators for religion, social group, wealth category, age of the household head, household size, and area of residence. Geographic characteristics include ruggedness, elevation, night lights. Standard errors (in parentheses) are clustered at the neighbourhood level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 4: Extension Approach:Disaggregated outcome variable

	Women's Period Product Use			
	Sanitary Napkins	Locally Prepared Napkins	Tampons	Menstrual Cups
	(1)	(2)	(3)	(4)
Age at Marraige	0.07*** (0.01)	0.10*** (0.02)	0.01 (0.01)	0.01** (0.00)
Observations	66,411	32,670	24,918	24,102
Woman's Characteristics	Yes	Yes	Yes	Yes
Household Characteristics	Yes	Yes	Yes	Yes
Geographical Characteristics	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Kleibergen-Paap rK-LM Statistic	243.66	111.42	114.75	120.48
First Stage F Statistic	262.93	114.95	122.80	129.60

The outcome variables are various products women use during periods. Women's characteristics include age, height and media exposure. Household characteristics include indicators for religion, social group, wealth category, age of the household head, household size, and area of residence. Geographic characteristics include ruggedness, elevation, night lights. Standard errors (in parentheses) are clustered at the neighbourhood level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 5: Extension Approach: Alternative Outcome Variable and Alternative Estimation Strategy

	(1)	(2)	(3)
Age at Marriage	0.13*** (0.02)	0.11*** (0.01)	0.08*** (0.01)
Observations	78,779	56,238	78,086
Woman's Characteristics	Yes	Yes	Yes
Household Characteristics	Yes	Yes	Yes
Geographical Characteristics	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Kleibergen-Paap rK-LM Statistic	241.09	200.65	241.09
First Stage F Statistic	255.59	212.75	255.59

Notes: Column 1 and 2 depicts the results for an alternative definition of the outcome variable . First, we define the outcome to equal one only if a woman reports exclusive use of period products, and zero if she reports using any non period or no period products. Under this specification, women who report mixed use are classified as non-users. Second, we adopt a more restrictive definition by excluding women who report using both period and non period materials from the sample. Column 3 shows us the result for an alternative estimation strategy. The Kleibergen–Paap rk LM statistic and the corresponding F statistic reported in column (3) are obtained from the IV–2SLS specification. Women's characteristics include age, height and media exposure. Household characteristics include indicators for religion, social group, wealth category, age of the household head, household size, and area of residence. Geographic characteristics include ruggedness, elevation, night lights. Standard errors (in parentheses) are clustered at the neighbourhood level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 6: Mechanism : Estimates of the effect of women's age at marriage on their years of education and decision making status.

	IV	OLS	IV	OLS
	Years of Education	Women's Period Product Use	Women's Decision Making Status	Women's Period Product Use
	(1)	(2)	(3)	(4)
Age at Marriage	1.21*** (0.11)		0.16** (0.08)	
Fitted Education		0.02*** (0.00)		
Fitted Decision Making				0.12*** (0.01)
Observations	78,779	78,779	11,589	11,589
Woman's Characteristics	No	Yes	Yes	Yes
Household Characteristics	No	Yes	Yes	Yes
Geographical Characteristics	No	No	Yes	Yes
District FE	No	No	No	Yes
Kleibergen-Paap rK-LM Statistic	241.09		36.01	
First Stage F Statistic	255.59		34.95	

Notes: Columns (1) and (3) report estimates from IV-TSLS, whereas columns (2) and (4) present results from OLS regression. Women's characteristics include age, height and media exposure. Household characteristics include indicators for religion, social group, wealth category, age of the household head, household size, and area of residence. Geographic characteristics include ruggedness, elevation, night lights. Standard errors (in parentheses) are clustered at the cluster level in columns (1) and (3). Standard errors are bootstrapped with 100 replications to avoid the generated regressor problem in columns (2) and (4). ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 7: Robustness analysis by trimming the extreme values of the endogenous and instrument variable as well as other sample restrictions

	Women's Period Product Use							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Age at Marriage	0.08*** (0.02)	0.11*** (0.03)	0.19*** (0.05)	0.08*** (0.02)	0.10*** (0.02)	0.10*** (0.03)	0.07*** (0.01)	0.07** (0.03)
Observations	75,616	67,810	56,461	75,008	60,286	51,037	55,409	23,370
Woman's Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographical Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rK-LM Statistic	203.13	106.27	43.89	170.52	75.09	32.38	200.30	40.05
First Stage F Statistic	214.11	109.23	44.24	174.74	75.43	32.19	213.66	41.26

Notes: Estimation via IV-TSLS estimation. Columns (1), (2), and (3) excludes the top as well as bottom 1%, 5%, and 10% of the endogenous variable : age at marriage, respectively. Columns (4), (5), and (6) excludes the top as well as bottom 1%, 5%, and 10% of the instrumental variable : age at menarche, respectively. In Column (7) and column (8) we restrict our sample to observations collected prior to the onset of pandemic-related restrictions (till March 2020) and after the onset of pandemic-related restrictions (from April 2020) respectively. All specifications include women's characteristics, household characteristics, district-level socioeconomic and geographic controls, and state fixed effects. Women's characteristics include age, height and media exposure. Household characteristics include indicators for religion, social group, wealth category, age of the household head, household size, and area of residence. Geographic characteristics include ruggedness, elevation, night lights. Standard errors (in parentheses) are clustered at the neighbourhood level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 8: Robustness analysis by adding additional controls

	Women's Period Product Use	
	(1)	(2)
Age at Menarche	0.08** (0.03)	0.07* (0.04)
Observations	11,589	7,918
Woman's Characteristics	Yes	Yes
Household Characteristics	Yes	Yes
Geographical Characteristics	Yes	Yes
District FE	Yes	Yes
Husband Characteristics	Yes	Yes
Parental Characteristics	No	Yes
Kleibergen-Paap rK-LM Statistic	36.72	23.45
First Stage F Statistic	35.65	22.04

Notes: Estimation via IV-TSLS estimation. Columns (1), (2) include husband and parental characteristics respectively. All specifications include women's characteristics, household characteristics, geographic characteristics, and district fixed effects. Women's characteristics include age, height and media exposure. Household characteristics include indicators for religion, social group, wealth category, age of the household head, household size, and area of residence. Geographic characteristics include ruggedness, elevation, night lights. Husband characteristics include their age, years of education as well as employment in last 12 months. The parental characteristics include women's exposure to interparental violence. Standard errors (in parentheses) are clustered at the neighbourhood level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Appendix

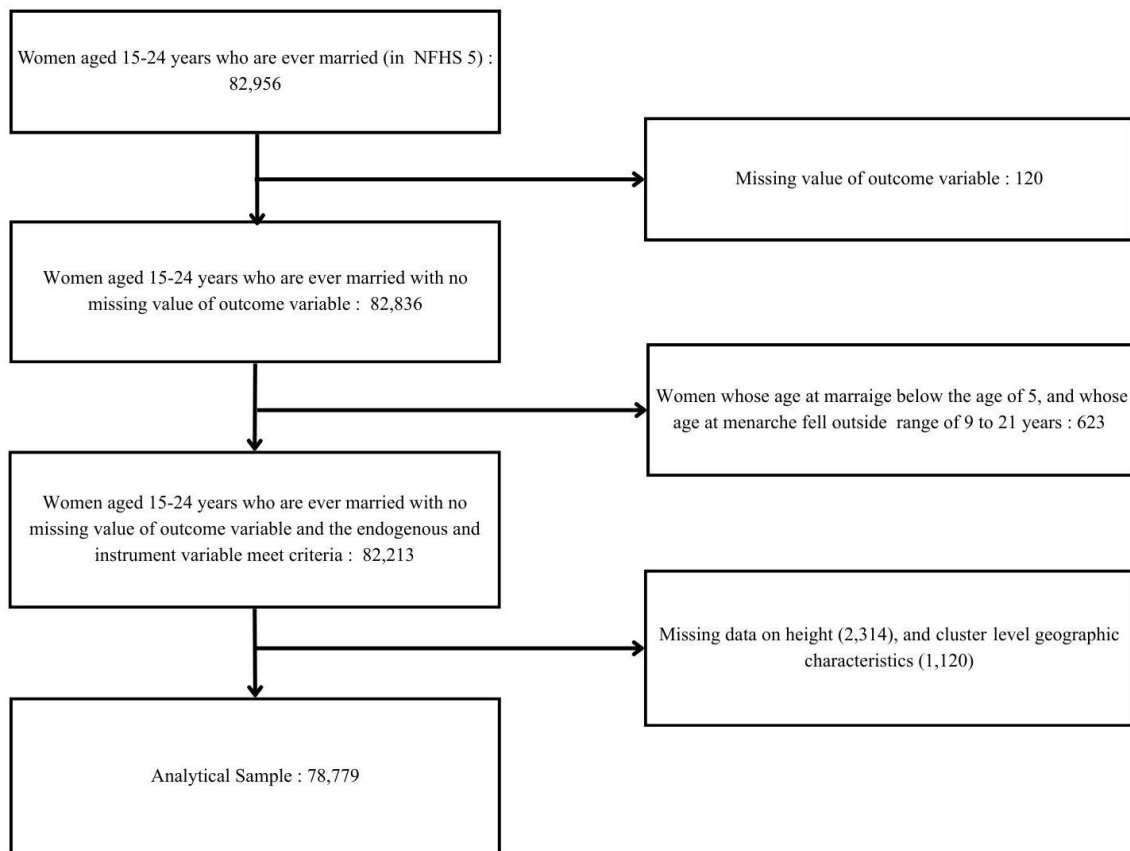


Figure A1: Analytical Sample formation